Multi-Resolution Semantic Abstraction over an Evolving Knowledge Hypergraph

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Abstract

Scientific knowledge grows continuously. New research papers introduce new methods, datasets, techniques, and relationships between existing concepts. Because of this growth, large knowledge structures become difficult to analyse directly.

This work studies a Temporal Knowledge Hypergraph (TKH) and develops a multi-resolution semantic abstraction framework. The objective is to create hierarchical representations of scientific knowledge while preserving semantic meaning, higher-order relationships, and temporal evolution.

The proposed pipeline contains temporal snapshot construction, semantic representation, a hypergraph-based objective, multi-resolution hierarchy construction, temporal coupling, semantic labelling, and evaluation.

The final system produces an interpretable hierarchy over an evolving knowledge hypergraph. The hierarchy is evaluated through intrinsic analysis and a downstream retrieval task.

1 Introduction

Scientific knowledge is not static. Every year, new scientific papers introduce new methods, datasets, and connections between concepts.

Many knowledge systems represent information as normal graphs, where each relationship connects two entities. However, scientific relationships are often higher-order.

For example, one scientific paper can:

- introduce a new method;
- use several datasets;
- apply multiple techniques;
- solve several scientific problems.

A pairwise graph representation can lose this information.

For this reason, this project uses a Temporal Knowledge Hypergraph (TKH). A hypergraph allows one relationship to connect multiple entities at the same time.

The main goal of this project is to create a multi-resolution representation of an evolving scientific knowledge structure.

The central research question is:

> How can an evolving knowledge hypergraph be transformed into an interpretable hierarchy while preserving semantic information, higher-order relationships, and temporal changes?

The proposed framework creates several abstraction levels. Lower levels contain more detailed entities, while higher levels represent more general scientific concepts.

The complete workflow is shown in Figure 1.

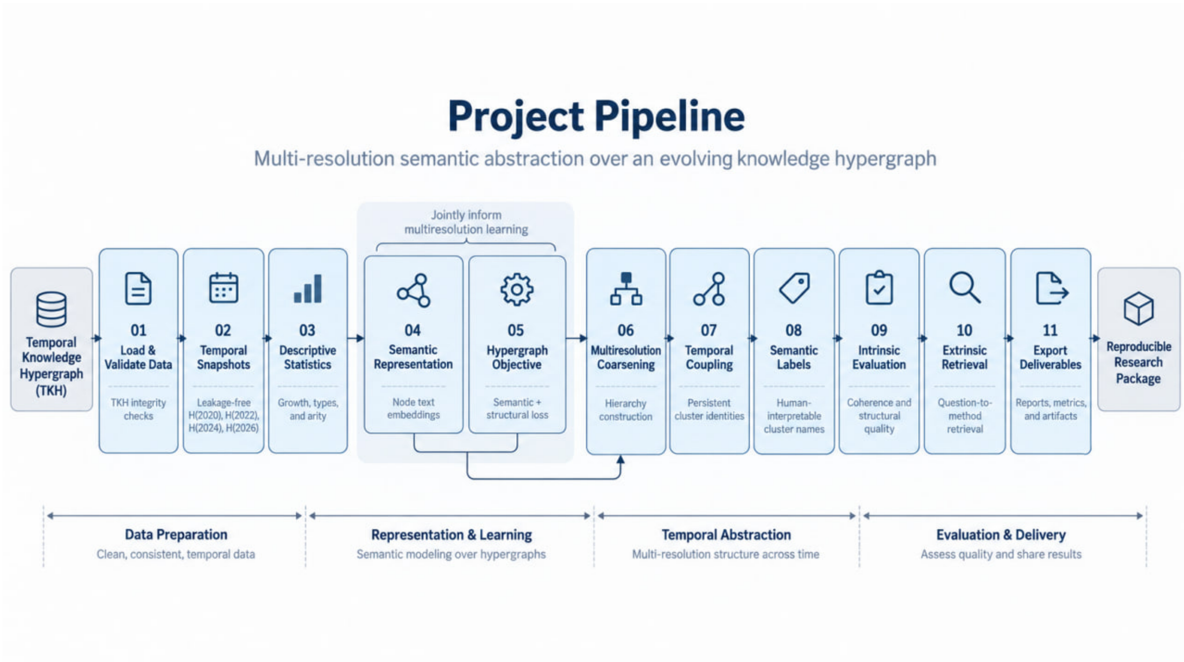

Figure 1: Complete workflow of the multi-resolution semantic abstraction pipeline.

2 Problem Statement

The input of this project is a Temporal Knowledge Hypergraph:

$$H(t) = (V(t), E(t))$$

where:

- $V(t)$ represents the knowledge entities available at time t;
- $E(t)$ represents higher-order relationships between multiple entities.

The objective is to construct a hierarchy:

$$P_0, P_1, ..., P_K$$

where each level provides a different resolution of the knowledge structure.

The hierarchy should satisfy three main requirements.

2.1 Semantic consistency

Entities grouped together should have related meanings.

For example, scientific methods that solve similar problems should appear in the same or nearby abstract groups.

2.2 Hypergraph preservation

The original TKH contains higher-order relationships.

The abstraction should preserve these relationships instead of converting the problem into a simple pairwise graph.

2.3 Temporal consistency

The knowledge structure changes over time.

The same scientific concept should remain traceable between different temporal snapshots.

Therefore, the final hierarchy should not only describe the current TKH but also represent how scientific knowledge changes.

3 Dataset and Temporal Hypergraph Representation

The project uses the Temporal Knowledge Hypergraph provided for the assessment task.

The first stage validates the original data structure.

The validation checks:

- node identifiers;
- hyperedge identifiers;
- node references;
- temporal information;
- semantic descriptions;
- provenance information.

After validation, the TKH contains:

Property	Value
Number of nodes	5798
Number of hyperedges	1429
Temporal snapshots	2020, 2022, 2024, 2026

Table 1: Basic statistics of the validated Temporal Knowledge Hypergraph.

The hypergraph contains heterogeneous scientific entities, including:

- methods;
- techniques;
- datasets;
- components;
- tasks;

- problems;
- articles.

Unlike a normal graph, hyperedges can contain multiple entities.
The hyperedge size statistics are:

Statistic	Value
Minimum hyperedge size	2
Maximum hyperedge size	65
Mean hyperedge size	6.25
Median hyperedge size	3
Multi-node hyperedge ratio	0.805

Table 2: Hyperedge statistics of the TKH.

4 Temporal Snapshot Construction

The TKH evolves over time because new entities and relationships appear.
Instead of using one static graph, this project creates several historical snapshots:

$$H(2020), H(2022), H(2024), H(2026)$$

A node is included in a snapshot only if it was already known at that time:

$$first\_seen\_year(v) \leq t$$

This prevents future information from appearing in earlier snapshots.
For a hyperedge:

$$e = (v_1, v_2, ..., v_n)$$

the hyperedge is included only when all of its members exist in the snapshot:

$$v_i \in V_t \quad \forall v_i \in e$$

The original hyperedge is not shortened. This keeps the original meaning of higher-order relationships.
The generated temporal snapshots are:

Year	Nodes	Hyperedges
2020	1505	374
2022	2164	529
2024	4164	983
2026	5798	1429

Table 3: Temporal snapshot statistics.

4.1 Semantic Representation

Each TKH entity contains textual information such as names, descriptions, and scientific metadata.

To compare entities based on meaning, each entity is transformed into a semantic vector:

$$x_v \in R^d$$

where d is the embedding dimension.

The complete semantic representation is:

$$X_{sem} \in R^{|V| \times d}$$

where:

- $|V|$ is the number of entities;
- d is the dimension of the semantic embedding.

The semantic representation provides information about the meaning of entities. It allows the system to identify related scientific concepts even when their surface names are different.

For example, two methods may have different names but belong to the same scientific area.

The semantic representation is created independently from benchmark questions and evaluation labels. Therefore, the retrieval evaluation does not use future information.

5 Hypergraph-Native Objective

The main challenge of abstraction is to create meaningful groups without destroying the original hypergraph structure.

For this reason, the abstraction objective combines two components:

1. semantic coherence;
2. hypergraph structural preservation.

The objective function is:

$$J_\alpha(P) = \alpha L_{sem}(P) + (1 - \alpha) L_{hyp}(P)$$

where:

- P is a hierarchy partition;
- L_{sem} measures semantic inconsistency;
- L_{hyp} measures hyperedge fragmentation;
- α controls the importance of each component.

5.1 Semantic coherence term

The semantic component encourages related entities to be placed together.

For a cluster C:

$$L_{sem}(C) = \frac{1}{|C|} \sum_{v \in C} d(x_v, \mu_C)$$

where:

- x_v is the semantic vector of node v;
- μ_C is the semantic center of cluster C;
- d is a distance function.

A smaller value means that the cluster contains semantically similar entities.

5.2 Hypergraph preservation term

The TKH contains higher-order relationships.

A hyperedge is represented as:

$$e = (v_1, v_2, ..., v_n)$$

After abstraction, the members of a hyperedge may be distributed between different clusters.

For a hyperedge e, the proportion of members assigned to cluster c is:

$$p_{e,c} = \frac{|e \cap C_c|}{|e|}$$

The fragmentation of the hyperedge is measured using entropy:

$$h_e(P) = -\sum_c p_{e,c} \log(p_{e,c})$$

Low entropy means that most members of the hyperedge remain inside the same abstract group.

High entropy means that the relationship is divided across multiple groups.

The final structural loss is:

$$L_{hyp}(P) = \frac{1}{|E|} \sum_{e \in E} h_e(P)$$

This objective keeps the original higher-order nature of the TKH.

The method does not first convert the hypergraph into a pairwise graph.

6 Multi-Resolution Hierarchy Construction

The goal of the abstraction stage is to create several views of the knowledge structure.

The hierarchy is represented as:

$$Level_0 \rightarrow Level_1 \rightarrow ... \rightarrow Level_K$$

where:

- lower levels contain detailed entities;
- higher levels contain more general concepts.

Each abstract node stores its relationship with the original TKH entities.
Therefore, users can move between:

$$\text{abstract concept} \rightarrow \text{original scientific entities}$$

This traceability is important because abstraction should not remove the connection with the original knowledge.

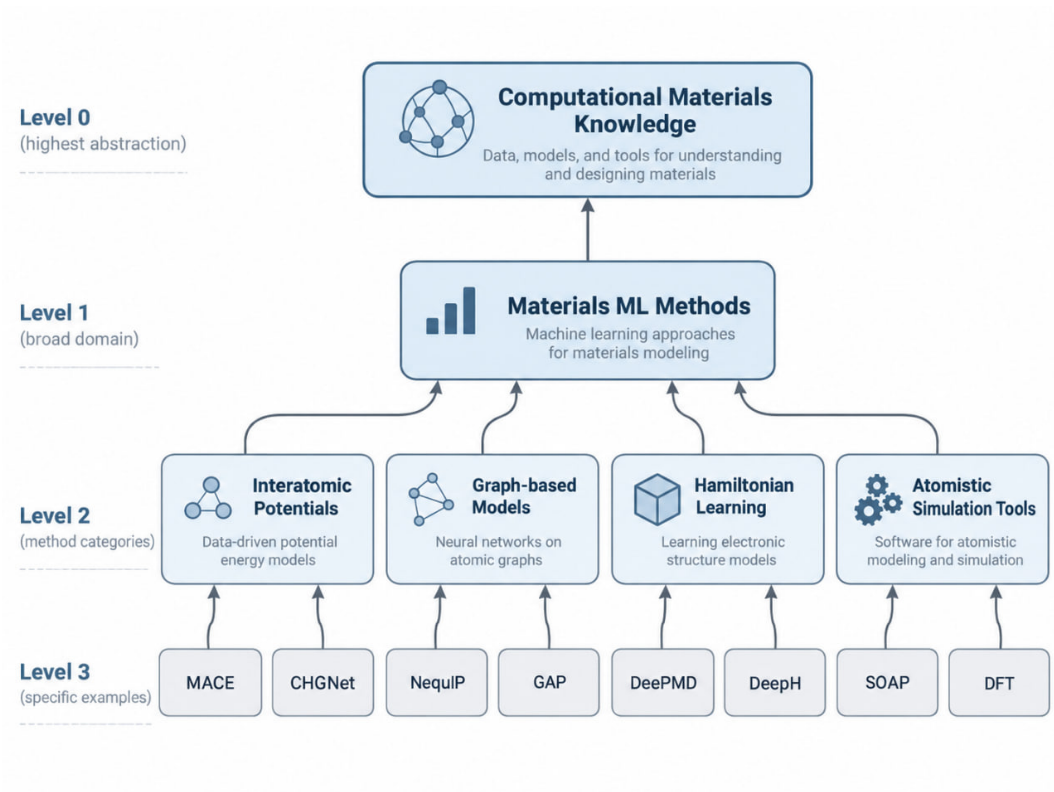

Figure 2: Example of the multi-resolution hierarchy representation.

The final hierarchy contains multiple abstraction levels, allowing analysis at different resolutions.

7 Temporal Coupling

A major challenge in temporal hierarchy construction is maintaining identity over time.

If each snapshot is processed independently, the same scientific concept may receive different identifiers:

$$C_1^{2020} \neq C_5^{2022}$$

even if both represent the same underlying concept.

To solve this problem, the project introduces temporal coupling between hierarchy snapshots.

The objective is to connect abstract entities between consecutive timestamps.
The evolution of a cluster can be represented by events:

- continuation;
- growth;
- split;
- merge;
- disappearance.

For example:

$$C_{2020} \to C_{2022} \to C_{2024} \to C_{2026}$$

represents the evolution of one abstract scientific concept over time.

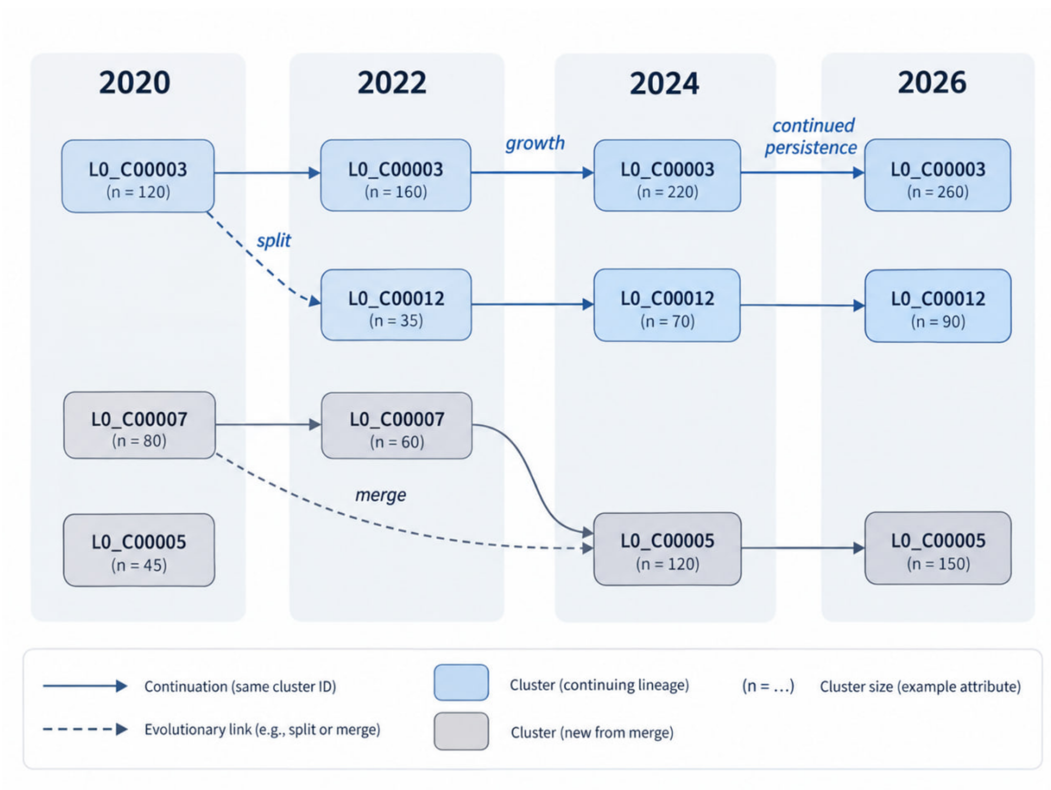

Figure 3: Temporal coupling connects hierarchy entities across different snapshots.

Temporal coupling transforms independent static hierarchies into an evolving knowledge representation.

8 Semantic Labelling

The final hierarchy should be understandable by human users.
Therefore, each abstract entity receives:

- a semantic label;
- a short description;
- links to its member entities.

The labels are generated from the information contained in the hierarchy members.

The purpose of labelling is not only visualization, but also supporting downstream retrieval and interpretation.

A labelled abstract node contains:

$$Cluster = \{label, gloss, member\_entities\}$$

where the member entities preserve the connection with the original TKH.

9 Evaluation Methodology

The evaluation has two purposes.

First, we evaluate whether the generated hierarchy is meaningful internally.

Second, we evaluate whether the hierarchy can support downstream scientific retrieval.

Therefore, the evaluation contains two parts:

1. Intrinsic evaluation.
2. Extrinsic retrieval evaluation.

9.1 Intrinsic Evaluation

Intrinsic evaluation measures the quality of the hierarchy without using external benchmark questions.

The evaluation focuses on three properties.

9.1.1 Semantic coherence

A good hierarchy should group entities with similar meanings.

If unrelated entities are grouped together, the abstraction quality decreases.

9.1.2 Hypergraph preservation

The abstraction should preserve the original higher-order relationships.

The evaluation considers whether hyperedges remain concentrated inside abstract groups or become strongly fragmented.

9.1.3 Temporal consistency

The hierarchy should evolve smoothly over time.

The temporal coupling mechanism is evaluated by checking whether abstract entities maintain persistent identities across snapshots.

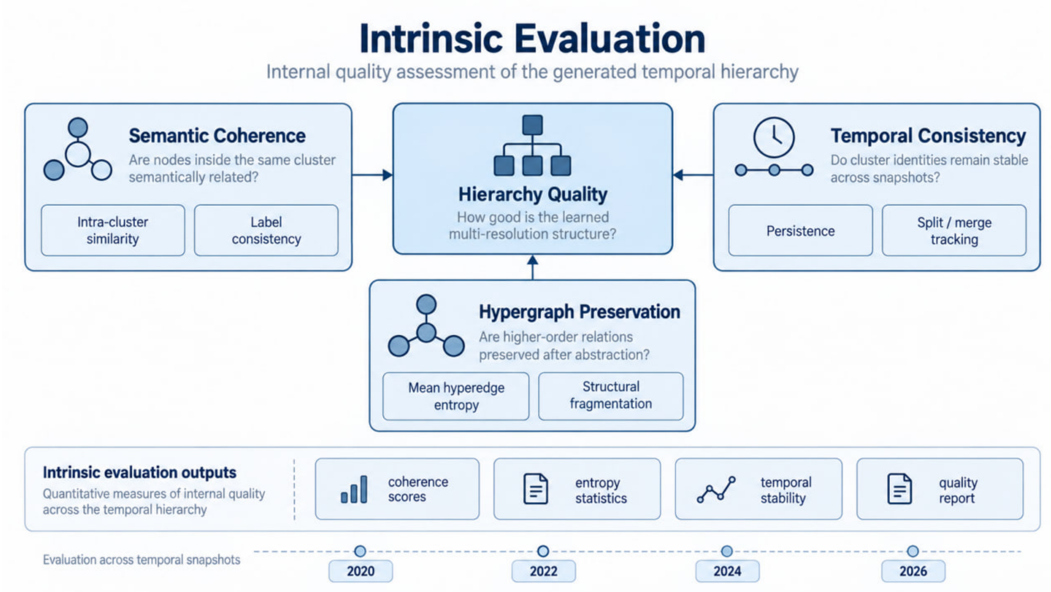

Figure 4: Intrinsic evaluation dimensions for the generated hierarchy.

9.2 Extrinsic Retrieval Evaluation

The second evaluation tests whether the generated hierarchy is useful for scientific information retrieval.

The benchmark contains questions with expected scientific methods.

The retrieval process is:

$$Question \rightarrow SemanticRepresentation \rightarrow SimilaritySearch \rightarrow RetrievedConcepts$$

The retrieved concepts are compared with the expected methods.

9.2.1 Evaluation Metrics

The evaluation uses precision, recall, and F1 score:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2\frac{Precision \times Recall}{Precision + Recall}$$

The current evaluation uses normalized exact matching between benchmark names and retrieved labels.

10 Results

10.1 Temporal Growth Results

The temporal snapshots show continuous growth of the knowledge hypergraph.

Year	Nodes	Hyperedges
2020	1505	374
2022	2164	529
2024	4164	983
2026	5798	1429

Table 4: Growth of the Temporal Knowledge Hypergraph.

The results show that the TKH continuously expands as new scientific knowledge is added.

10.2 Hierarchy Results

The generated hierarchy provides several abstraction levels.

Each abstract entity maintains links to its original members.

This allows users to move between:

$$\text{high-level concept} \rightarrow \text{original scientific entities}$$

The hierarchy therefore provides both abstraction and traceability.

10.3 Retrieval Results

The retrieval evaluation produced low exact-match scores.

The reason is not only retrieval quality.

A major challenge is the difference between benchmark terminology and TKH terminology.

The benchmark usually contains abbreviated method names:

```
MACE

GAP

DeepH

xDeepH

DeePMD
```

However, TKH entities often contain complete names or scientific descriptions:

```
Machine-learned interatomic potentials
```

```
Universal machine learning interatomic potentials

deep-learning DFT Hamiltonian

Deep Potential Molecular Dynamics
```

Therefore, two concepts can describe the same scientific idea but still fail an exact string comparison.

10.4 Retrieval Case Analysis

To better understand the retrieval behaviour, several representative questions are analysed manually.

ID	Benchmark question	Expected methods	Retrieved concepts
Q1	Which methods (by Feb 2026) are best suited for predicting interatomic potentials when balancing accuracy close to DFT with computational efficiency for systems of millions of atoms?	MACE, GAP, M3GNet, NequIP, CHGNet	Machine-learned interatomic potentials; Universal machine learning interatomic potentials
Q2	Which methods (by Feb 2026) are best suited for accelerating electronic structure calculations when self-consistent field iterations become prohibitively expensive for large systems ($> 10^3$ atoms)?	D4FT, DeepH, HamGNN	AI-accelerated electronic structure prediction concepts
Q3	Which methods (by Feb 2026) are best suited for learning DFT Hamiltonians when dealing with twisted van der Waals heterostructures containing thousands of magnetic atoms?	DeepH, xDeepH	deep-learning DFT Hamiltonian; deep neural network approach to represent DFT Hamiltonian
Q4	Which methods (by Feb 2026) are best suited for phonon property prediction when universal MLIPs show inaccuracies despite good energy/force performance?	VGNN, MACE-F, Hessian training	MLIP evaluation framework; Linear response phonon method
Q5	Which methods (by Feb 2026) are best suited for coarse-graining molecular dynamics when crystalline symmetries and long-range elastic interactions must be preserved?	Bead-mapping, GNN+GPP	Multiscale modelling and machine-learning related concepts

Table 5: Examples of semantic similarity between benchmark questions, expected methods, and retrieved TKH concepts. Exact matching fails when abbreviated benchmark terms and full TKH descriptions refer to the same scientific concepts.

For example, in Q1 the benchmark expects:

```
MACE, GAP, M3GNet, NequIP, CHGNet
```

The retrieved concepts are:

```
Machine-learned interatomic potentials
```

```
Universal machine learning interatomic potentials
```

The exact comparison gives no match.

However, both retrieved concepts represent the same scientific research area: machine learning interatomic potentials.

Similarly, for Q3:

Expected:

```
DeepH, xDeepH
```

Retrieved:

```
deep-learning DFT Hamiltonian

deep neural network approach to represent DFT Hamiltonian
```

Again, the retrieved concepts are semantically related but use different surface forms.

11 Discussion and Limitations

The project demonstrates that a temporal knowledge hypergraph can be converted into an interpretable multi-resolution hierarchy.

The main contributions are:

1. Temporal snapshots that avoid future information leakage.
2. A hypergraph-native objective that preserves higher-order relationships.
3. A multi-resolution hierarchy with traceability to original entities.
4. Temporal coupling that connects knowledge evolution across time.
5. A retrieval evaluation pipeline for downstream testing.

However, several limitations remain.

11.1 Vocabulary mismatch in retrieval evaluation

The main limitation appears in the extrinsic evaluation.

The benchmark and TKH use different naming styles.

The benchmark frequently uses short names:

`MACE`

`GAP`

`DeepH`

while TKH often stores expanded scientific descriptions.

Because the current metric uses exact matching, semantic matches are not always counted as correct.

Future work should introduce a stronger entity linking step using:

- scientific aliases;
- ontology matching;
- embedding-based semantic matching;
- expert validation.

11.2 Semantic representation limitation

The quality of abstraction depends on the available textual information.

Entities with limited descriptions may receive weaker semantic representations.

11.3 Hierarchy evaluation limitation

Large hierarchical structures are difficult to evaluate completely using only automatic metrics.

Future work could include expert evaluation of generated abstract concepts.

12 Conclusion

This work presents a framework for multi-resolution semantic abstraction over an evolving knowledge hypergraph.

The proposed pipeline combines:

- temporal snapshot construction;
- semantic representation;
- hypergraph structural preservation;
- multi-resolution hierarchy construction;
- temporal coupling;
- semantic labelling;
- intrinsic and extrinsic evaluation.

The final system transforms a large temporal knowledge hypergraph into an interpretable hierarchy while preserving links to the original scientific entities.

The retrieval evaluation shows that the hierarchy can identify related scientific concepts. However, exact benchmark matching is limited by naming differences between abbreviated benchmark terms and full TKH descriptions.

Future improvements should focus on stronger entity linking and semantic evaluation methods.

Acknowledgements

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